Tushar Anand

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Experience

Clickpe (YC W23) **Remote**

AI Engineer Intern *January 2026 - July 2026*

- Built **AI based voice calling** infrastructure to **handle 5k+** incoming and outgoing calls to customers on daily basis.
- Orchestrated an **Agentic pipeline** with **twilio and tatatele** for telephony and stt models form **Deepgram**,llm from **cerebras and groq**, tts models form **caretessia** with **pipecat** and achieved **sub 800ms** latency during calls.
- Reduced **tts** cost by **30%** using redis to **cache audio blobs** with hashing and **llm** costs by **60%** with **prefix caching**.
- Integrated **langfuse** for tracing along with **custom dashboards** to track various metrics and **detect hallucinaitons**.
- Created **n8n workflows** to automatically **schedule calls** and internal dashboard to **track status** of ongoing calls.
- Used **in-memory rag** to **insert dynamic information** in prompt based on user query during the call **reducing** system prompt **size by 40%.** and improving **inference** speed **by 10-15% on average**.

AsyncApi Mentee **Remote**

Open Source Contributor *July 2026 - September 2026*

- Contributed to the **AsyncAPI** projects sponsored by **RedHat and IBM**. Added new commands, enhancing existing ones, and improving test coverage for a reliable developmet. Cleaned up unused dependencies and reduced project build size across platforms.
- Implemented a new **preview command** enhancing the end user UX and additon of hedless browser testing with puppeteer for studio command in cli, for which a reward of $400 was awarded under the AsyncApi bounty program.
- Upgraded dependencies while **maintaining compatibility** across AsyncAPI templates to improve performance and maintainability along with improving build times by 40% and reducing docker size with multi stage builds.
- Contributed to the **AsyncApi Studio** poject by adding support for **local schema file** previewing with CLI.
- Got a total of **26 pull requests** merged across CLI and other repositories along with helping maintainers in review.

Creuto **Remote**

Software Development Intern *Jan 2025 - July 2025*

- Developed the backend for an internal application from scratch for **scheduling meetings** within the company, reducing the need for 3rd party applications and cost by **70%**.
- Integrated Razorpay for Payments and used Queues for a durable payments and payout processing architecture.
- Make My Look: Developed the backend for a salon booking application incorporating **Razorpay** for payments,**Firebase cloud messaging** for on device reltime notificaions and **MongoDB geospatial** queries for **location based service** recommendation and **redis** for caching popular services data handling over **1k+ users efficiently**.

Projects

Rumik-Fast – Real-time 3B TTS Engine on a 4 GB GPU | *CUDA, Triton, PyTorch*

- Built a from-scratch **inference engine** for a **3B Hindi/English TTS** model, raising decode speed from **59 to 98 tok/s** (real time) on a **4 GB RTX 3050** laptop GPU, close to its **memory-bandwidth limit**.
- Wrote **CUDA int4/int8 GEMV kernels** and fused **Triton** kernels with **flash-decoding**, all replayed in one **CUDA graph** with a static **KV cache**.
- Cut prefill from **642 to 50 ms** with an int4 **tensor-core** matmul and added **exact streaming** audio decoding: **first audio in ∼150 ms**.

Gemma-STT-Hindi – Speech-to-Text via Distillation | *PyTorch, KenLM, Kaggle*

- Trained a **Hindi speech-to-text** model on Google's **EmbeddingGemma 2** audio encoder with a **CTC** head, entirely on **free Kaggle GPUs**; released on **Hugging Face**.
- **Distilled** two larger ASR models by relabelling **1,750 h** of YouTube audio; showed **label quality beats hours** (780 h of teacher labels beat 1,810 h of captions).
- Built a **KenLM n-gram** for **beam-search** decoding, cutting WER by **∼25%**; reached **5.4% CER** on Lahaja, on par with **Qwen3-ASR-1.7B**, at **40 ms per clip**.

Technical Skills

Languages:C, Typescript, Python, SQL, Bash
Technologies: Git, Docker, Cloudflare, AWS, Postgres, Github Actions, Kubernetes, Linux

Education

International Institute of Information Technology **Bhubaneshwar, Odisha**

Bachelor of Technology in ***Computer Engineering*** ***Aug 2023 - Aug 2027***

GPA: 8.1/10.0

- **Coursework:** Data Structures And Algorithms, Statics, Computer Architecture and Digital Logic, Networking.